From Social Dilemmas to Artificial Societies: Measuring Baseline, Steerability, and Institutional Behavior in LLM Agents

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Abstract

Large language models are increasingly deployed as interactive agents, but their social behavior is often evaluated in ways that conflate default strategic tendencies, prompt obedience, benchmark familiarity, and institution-mediated adaptation. We present a behavioral evaluation framework that separates these factors across repeated classical games and larger multi-agent settings. The repaired empirical core now spans repeated Prisoner's Dilemma, Chicken, and Stag Hunt experiments plus corrected scarcity and public-reputation society reruns, all with full prompt-stack logging after a validity repair. We find five robust results. First, game structure strongly shapes behavior: on a repaired stable cohort, cooperation rises from Prisoner's Dilemma (`0.4722 / 0.5463`) to Chicken (`0.9074 / 0.7870`) to Stag Hunt (`0.9722 / 0.9907`), and the same ordering replicates on a second same-day cohort. Second, even neutral wording is not behaviorally innocuous: pooling two repaired Prisoner's Dilemma cohorts, `minimal-neutral` yields `0.2639 / 0.2917` cooperation versus `0.5833 / 0.6667` for two other neutral paraphrases. Third, prompt framing can steer the same game from universal defection to universal cooperation. Fourth, benchmark presentation materially changes measured behavior, but not in one direction across games: removing canonical framing raises cooperation in Prisoner's Dilemma while lowering it in Chicken and Stag Hunt. Fifth, institutional effects are not reducible to surface prosociality: in the corrected scarcity society, `task-only` preserves `1.0000` final survival while both `cooperative` and `competitive` fall to `0.8889`, whereas public reputation equalizes final survival at `1.0000` across all three prompt families while preserving large differences in trade, alliances, and inequality. These findings support a behavioral view of alignment in which baseline policy, steerability, benchmark recognition, and institutional mediation must be measured separately rather than collapsed into a single cooperation score.

1. 1. Introduction

Large language model alignment is often evaluated through refusal behavior, static benchmarks, or capability tests. Yet many realistic deployments are social: models negotiate, coordinate, compete, advise, and participate in institutional settings where long-horizon incentives matter. In these environments, alignment is partly a behavioral question. What does a model do when cooperation is jointly efficient but individually risky? How stable is that behavior under neutral paraphrase? Does it change when a famous benchmark label is removed? And can prosocial language be separated from robust collective outcomes?

This paper argues that LLM social behavior should not be summarized by a single scalar notion such as "cooperativeness" or "altruism." Instead, behavioral evaluation should distinguish at least four targets:

1. baseline behavior under minimally directive prompts
2. sensitivity to prompt framing
3. sensitivity to benchmark presentation and recognizability
4. adaptation to institutions such as scarcity and public reputation

That separation matters methodologically as well as substantively. If canonical benchmark names shift behavior, then some measured "strategy" may reflect recognition rather than default policy. If cooperative personas move outcomes, then we may be measuring instruction-following rather than

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stable social preferences. If public institutions change behavior, then those changes should be attributed to environmental structure rather than model essence.

The repaired empirical core in this manuscript centers on repeated classical games, where the evidence is deepest, and now includes a small corrected institutional battery under scarcity and public reputation. These repaired results already support a strong paper: default behavior is game-dependent, neutral wording is unstable, prompt framing is highly steerable, benchmark presentation materially changes measured cooperation, and institutional structure can separate visible social activity from collective survival.

2. 2. Related Work

Recent work shows that LLMs can exhibit strategic behavior in classical games, but the literature often stops short of separating baseline behavior, prompt steerability, and benchmark familiarity. Lore and Heydari (2024) provide one of the closest predecessors by showing that both game structure and contextual framing shape strategic behavior in LLMs. Akata et al. (2025) extend repeated game analysis to broader families of cooperation and coordination tasks and show stable differences across game families. Fontana et al. (2024) focus specifically on iterated Prisoner's Dilemma behavior.

Work on social preferences and human comparison is also relevant. Schmidt et al. (2024) study altruistic advice in Dictator and Ultimatum Game settings, while Capraro et al. (2025) benchmark how well LLMs predict human tradeoffs between self-interest and others' welfare. These papers are useful for grounding fairness and altruism claims, but they do not center repeated strategic interaction plus benchmark-disguise conditions.

On the multi-agent side, Park et al. (2023) introduced generative agents as a simulation architecture for believable social behavior. More recent work such as AgentSociety (Piao et al., 2025), Simulating Cooperative Prosocial Behavior with Multi-Agent LLMs (Sreedhar et al., 2025), ALYMPICS (Mao et al., 2024), and MultiAgentBench (Zhu et al., 2025) provides important context for multi-agent evaluation and simulation. Our closest novelty gap relative to this literature is not the generic claim that LLMs can act socially. It is the explicit decomposition of four measurement targets that are often conflated: neutral baseline behavior, prompt susceptibility, benchmark recognition, and institution-mediated behavior.

3. 3. Experimental Design

3.1. 3.1 Measurement targets

The project separates three related but distinct questions. Baseline measurement asks what models do under minimally directive prompts. Prompt susceptibility asks how much that behavior changes under cooperative or competitive framing. Benchmark-recognition measurement asks whether canonical game labels change behavior relative to unnamed or disguised isomorphs.

3.2. 3.2 Repaired Part 1 protocol

The repaired Part 1 experiments use a stable three-model cohort:

- `cerebras:llama3.1-8b`
- `nvidia:deepseek-ai/deepseek-v3.2`
- `nvidia:moonshotai/kimi-k2-instruct-0905`

The repaired replication cohort swaps in Qwen for the Kimi model. Each game is run over all cross-model pairings plus self-play, yielding six pairings per game. In the repaired fast batch, each trial lasts four rounds at temperature `0.0`, and each pairing is evaluated once per condition. For the repaired baseline track, each game is evaluated under three neutral prompt variants, yielding eighteen trials per game. For the repaired benchmark track, each presentation is evaluated over the same six pairings. For the repaired susceptibility track, each game is evaluated under `minimal-neutral`, `cooperative`, and `competitive` prompts, again yielding eighteen trials per game.

3.3. 3.3 Validity repair

The most important methodological repair is that experiment logs now preserve the full effective message stack through `messages_sent`, rather than logging only the final task prompt. This matters because the actual model input includes the system prompt, framing, persona, and history context. Claims about neutral behavior are only defendable if the full effective prompt is preserved.

The paper bundle also now de-duplicates interrupted retries correctly. Whole directory summaries and figures prefer the latest completed JSON artifact for each logical experiment name and exclude stale partial JSONL retries.

4. 4. Results

4.1. 4.1 Game structure strongly shapes baseline behavior

The repaired stable cohort produces a clear cross-game ordering. Aggregate cooperation is lowest in Prisoner's Dilemma, higher in Chicken, and highest in Stag Hunt:

The same qualitative ordering replicates on a second same-day cohort: Prisoner's Dilemma at `0.4815 / 0.5370`,

Table 1. Repaired cross-game baseline on the stable triplet cohort.

Game	Trials	Cooperation A	Cooperation B	Average Payoff A	Average Payoff B
Prisoner's Dilemma	18	0.4722	0.5463	2.2500	1.8796
Chicken	18	0.9074	0.7870	2.5093	2.9907
Stag Hunt	18	0.9722	0.9907	3.9630	3.9074

Chicken at `0.6852 / 0.6389`, and Stag Hunt at `0.9722 / 0.9907`. This makes the core claim stronger than a single-cohort result: strategic environment matters at least as much as any simple model-wide "prosociality" label.

4.2. 4.2 Even neutral wording is behaviorally unstable

The strongest repaired neutral-robustness result is in Prisoner's Dilemma. On the repaired stable cohort, `minimal-neutral` yields `0.2500 / 0.3056` cooperation, while both `minimal-neutral-compact` and `minimal-neutral-institutional` yield `0.5833 / 0.6667`. Pooling the stable and replication cohorts gives:

This result supports a strong methodological conclusion: there is no single "prompt-free" social baseline here. Baseline behavior is a distribution over defensible neutral phrasings, and papers that report one default cooperation estimate without robustness checks risk overstating precision.

4.3. 4.3 Prompt framing strongly steers behavior, but by different amounts across games

Prompt susceptibility is now finalized across the three repaired core games. Prisoner's Dilemma is the most dramatically steerable: `competitive` yields `0.0000 / 0.0000`, `cooperative` yields `1.0000 / 1.0000`, and `minimal-neutral` remains much lower at `0.2500 / 0.3056`. Chicken and Stag Hunt are also steerable, but they retain much higher residual cooperation under competitive framing:

The safe claim is therefore not simply that prompt framing matters. It is that prompt steerability is itself game-dependent.

4.4. 4.4 Benchmark presentation materially changes measured behavior

The repaired benchmark track is now complete across the three core games. Prisoner's Dilemma shows the most dramatic presentation effect. The canonical condition yields `0.2500 / 0.3056` cooperation, the resource-disguise condition raises that to `0.5833 / 0.6667`, and the unnamed/isomorphic condition raises it further to `0.7500 / 0.8056`. Relative to canonical, the unnamed condition increases cooperation by `+0.5000 / +0.5000`.

Chicken and Stag Hunt move in the opposite direction. In Chicken, canonical framing yields `0.9167 / 0.7778`, while unnamed framing falls to `0.4444 / 0.5833`. In Stag Hunt, canonical framing yields `0.9167 / 0.9722`, while unnamed framing falls to `0.6667 / 0.8333`.

This pattern is theoretically important. Benchmark recognition is a real behavioral confound, but not a monotonic one. Less canonical presentation does not always increase cooperation. Instead, benchmark framing appears to interact with the strategic schema a model activates for a particular game.

4.5. 4.5 Multi-agent institutions reveal survival-sociality dissociations

The corrected multi-agent reruns now complete all three prompt conditions for both the scarcity society and the public-reputation society. In the scarcity society, `task-only` is perfectly stable across three repetitions (`1.0000` final survival, zero trade, zero alliances). Both non-task prompts reduce final survival to `0.8889 [0.8333, 1.0000]`, but in qualitatively different ways. `Cooperative` produces the most social and unequal regime, with average trade volume `2.1667`, alliance count `2.0000`, average gini `0.3253`, and commons health `0.5321`. `Competitive` remains alliance-free and much less unequal, with average trade volume `1.7083`, alliance count `0.0000`, average gini `0.1429`, and commons health `0.3767`.

In the reputation society, by contrast, all three prompt families preserve `1.0000` survival and final survival. Yet their social styles remain sharply separated. `Task-only` is nearly inactive (`0.0000` trade, `0.0000` alliances), `competitive` stays austere (`0.5000` trade, `0.0000` alliances, `0.2960` commons health), and `cooperative` yields the most active regime (`1.6667` trade, `2.3333` alliances, `0.6128` commons health, `0.2724` gini). The repaired institutional battery therefore supports a stronger claim than the earlier pilot: public reputation appears to buffer survival penalties without erasing prompt-induced differences in social organization.

5. 5. Discussion

These repaired results support three broader claims.

First, LLM social behavior is highly context-sensitive. The same models move substantially across games, prompt fram-

Table 2. Pooled repaired Prisoner's Dilemma neutral-variant replication across two cohorts.

Prompt Variant	Pooled Trials	Cooperation A	Cooperation B	Average Payoff A	Average Payoff B
minimal-neutral	12	0.2639	0.2917	1.6806	1.5417
minimal-neutral-compact	12	0.5833	0.6667	2.5000	2.0833
minimal-neutral-institutional	12	0.5833	0.6667	2.5000	2.0833

Table 3. Repaired prompt susceptibility across the three core games.

Game	Prompt Variant	Trials	Cooperation A	Cooperation B
Prisoner's Dilemma	competitive	6	0.0000	0.0000
Prisoner's Dilemma	cooperative	6	1.0000	1.0000
Chicken	competitive	6	0.6389	0.3333
Chicken	cooperative	6	1.0000	1.0000
Stag Hunt	competitive	6	0.4167	0.3056
Stag Hunt	cooperative	6	1.0000	1.0000

ings, and benchmark presentations. Second, neutral baseline measurement is fragile enough that it should be treated as a robustness problem, not a single number. Third, benchmark recognition deserves to be treated as a first-class methodological issue in LLM social evaluation. Canonical task names can change behavior enough to alter the substantive interpretation of a result.

Together, these findings argue for a behavioral evaluation agenda that decomposes baseline policy, steerability, recognizability, and institutional mediation rather than collapsing them. A model can look cooperative in one canonical benchmark, yet defect under a different neutral paraphrase, become far more cooperative once the benchmark label is removed, or sustain visibly cooperative behavior only under public accountability. Those are not minor nuisance effects. They change what the experiment is measuring.

6. 6. Limitations

This manuscript has four important limitations.

First, the strongest repaired claims still come from Part 1 rather than the full micro-to-macro program. Second, the repaired baselines use a small but stable model cohort chosen by provider access and action-readiness constraints. Third, the repaired runs use fast-batch settings, which improves throughput but limits the number of rounds and repetitions per condition. Fourth, the corrected institutional battery is still small, with only three repetitions per prompt family in each institution.

These limitations do not undercut the main contribution. The Part 1 repaired bundle already supports a publishable methodological and empirical paper on baseline instability, prompt steerability, and benchmark recognition in LLM social dilemmas, and the corrected institutional reruns now add a closed three-prompt contrast under scarcity and public reputation. They do, however, bound the precision of institution-level claims until the same patterns are replicated at larger scale.

7. 7. Reproducibility And Release Considerations

This paper is designed to be reproducible from code and artifacts rather than from narrative description alone. Experiments are config-driven, prompts live in versioned files, raw outputs are preserved in JSON or JSONL artifacts, and paper figures and summaries are regenerated by dedicated scripts. The paper bundle therefore points directly to result directories, pooled summaries, figure manifests, and regression tests rather than relying on hand-copied tables. The associated appendix lists the exact commands used to regenerate the current manuscript-facing summaries and figures.

One caveat is that provider availability is date-specific. The repaired model panel reported here was selected from models that were both accessible and experiment-ready during the April 2026 runs. Future reruns may therefore need to substitute models if provider catalogs, rate limits, or API behavior change. We treat that as an external source of variance to document, not a reason to obscure the original cohort.

The release posture should also avoid anthropomorphic overclaiming. These results do not show that a model is intrinsically altruistic, selfish, moral, or aligned in any deep sense. They show that behavior in social settings is measurably shaped by game structure, prompt framing, benchmark recognition, and institutional rules. Releasing prompts, logs, and code is therefore useful for scientific scrutiny, but the surrounding paper should consistently frame the findings as behavioral evidence under specified conditions rather than as evidence of stable inner motives.

8. 8. Conclusion

The repaired empirical core shows that LLM social behavior cannot be summarized by a single cooperation score. It depends jointly on strategic environment, neutral-prompt robustness, prompt steerability, and benchmark presentation. The resulting picture is behaviorally richer and methodologically less naive than a single canonical-game evaluation. For alignment research, that is the central lesson: before asking whether models are cooperative, altruistic, or aligned in the abstract, we must first ask which behavior we are measuring, under which prompt, in which game, and under what institutional frame.

This appendix collects the manuscript-adjacent details needed to reproduce the current paper bundle from the repository.

A. A. Artifact Map

Primary manuscript artifacts:

- `paper/MANUSCRIPT.md`
- `paper/RESULTS_TABLES.md`
- `paper/FIGURE_CAPTIONS.md`
- `paper/REFERENCES.md`
- `paper/ARTIFACT_INDEX.md`

Primary result summaries:

- `results/paper_ready_baseline_triplet/interim_summary.md`
- `results/paper_ready_replications/interim_summary.md`
- `results/paper_ready_benchmark_triplet/interim_summary.md`
- `results/paper_ready_susceptibility_triplet/interim_summary.md`
- `results/paper_ready_society_triplet/interim_summary.md`
- `results/paper_ready_reputation_triplet/interim_summary.md`

Primary paper figures:

- `paper/figures/repaired_pd_replications/baseline_prompt_variants_cooperation.png`
- `paper/figures/triplet_live/baseline_prompt_variants_cooperation.png`
- `paper/figures/benchmark_live/benchmark_presentations_cooperation.png`
- `paper/figures/susceptibility_live/susceptibility_prompt_variants_cooperation.png`
- `paper/figures/society_reputation_live/society_reputation_final_survival.png`
- `paper/figures/society_reputation_live/society_reputation_trade_volume.png`
- `paper/figures/society_reputation_live/society_reputation_alliance_count.png`

B. B. Repaired Experimental Settings

B.1. B.1 Part 1 repeated-game cohort

Stable repaired cohort:

- `cerebras:llama3.1-8b`
- `nvidia:deepseek-ai/deepseek-v3.2`
- `nvidia:moonshotai/kimi-k2-instruct-0905`

Same-day replication cohort:

- `cerebras:llama3.1-8b`
- `nvidia:deepseek-ai/deepseek-v3.2`
- `cerebras:qwen-3-235b-a22b-instruct-2507`

Protocol highlights:

- self-play plus all cross-model pairings
- four repeated rounds per repaired fast-batch trial
- temperature `0.0`
- full `messages_sent` prompt-stack logging

B.2. B.2 Part 2 scarcity society

Population:

- two agents per model from the stable repaired triplet cohort

World settings:

- initial public resources `36`
- maximum public resources `48`
- regeneration rate `0.2`
- initial agent resources `6`
- gather amount `3`
- steal amount `2`
- survival cost `1`

- reproduction threshold `15`
- offspring start resources `4`
- maximum agents `12`

Institutional settings:

- private messages enabled
- stealing enabled
- unmonitored agents enabled
- unmonitored fraction `0.2`
- trade-offer TTL `3`

B.3. B.3 Part 3 public reputation society

Part 3 uses the same world configuration as Part 2, with:

- public reputation enabled
- reputation decay `0.95`
- anonymous ratings disabled
- rating range `1` to `5`
- no unmonitored agents

C. C. Metrics

C.1. C.1 Repeated-game metrics

- cooperation by agent position
- average payoff by agent position
- benchmark-presentation deltas
- prompt-susceptibility deltas

C.2. C.2 Society metrics

- `survival_rate`
- `final_survival_rate`
- `average_trade_volume`
- `average_gini`
- `commons_health`
- `alliance_count`
- `extinction_event`
- `commons_depletion_rate`

C.3. C.3 Reporting convention

Paper summaries report pooled means and 95% confidence intervals where available. Whole-directory aggregation prefers the latest completed artifact for each logical experiment name and excludes stale interrupted JSONL retries.

D. D. Reproduction Commands

All commands below were run from the repository root.

Regenerate corrected institutional summaries:

```
.venv/bin/python scripts/paper_summary.py
    results/paper_ready_society_triplet \
  --markdown results/
      paper_ready_society_triplet/
      interim_summary.md \
  --csv results/paper_ready_society_triplet/
      interim_summary.csv

.venv/bin/python scripts/paper_summary.py
    results/paper_ready_reputation_triplet \

  --markdown results/
      paper_ready_reputation_triplet/
      interim_summary.md \
  --csv results/
      paper_ready_reputation_triplet/
      interim_summary.csv
```

Regenerate manuscript-facing society/reputation figures:

```
.venv/bin/python scripts/paper_figures.py \
  results/paper_ready_society_triplet \
  results/paper_ready_reputation_triplet \
  --output-dir paper/figures/
      society_reputation_live
```

Build the anonymous ICML-style submission PDF:

```
.venv/bin/python scripts/
    build_icml_submission.py
```

Run paper-related regression coverage:

```
.venv/bin/python -m pytest -q \
  tests/test_paper_summary.py \
  tests/test_paper_figures.py \
  tests/test_paper_batch.py \
  tests/test_experiments.py
```

E. E. Threats To Validity

The most important repaired validity issue was incomplete prompt logging. Older artifacts that logged only the final task prompt should not be treated as the paper record when repaired artifacts exist with full message-stack logging.

Provider availability is also unstable over time. The accessible model panel is therefore a dated empirical object, not a timeless canonical benchmark. This paper addresses that by

documenting the exact repaired cohorts rather than claiming they are exhaustive.

Finally, the institutional battery remains modest in scale. The corrected Part 2 and Part 3 reruns are now complete across all three prompt families, but only at three repetitions per condition. That is sufficient for a defensible behavioral result set, but not for overconfident claims about universal institutional laws.

F. F. Ethics And Release

This project should be released as a behavioral benchmark and artifact bundle, not as evidence that models possess stable moral traits. Terms such as ”cooperative,” ”competitive,” or ”aligned” should always be interpreted as conditioned behavioral descriptors under specific prompts, games, and institutions.

Prompt files, configs, and logs are appropriate to release because they enable auditing and replication. The main ethical risk is misinterpretation: anthropomorphic readings can make narrow behavioral findings sound like claims about intention or character. The paper should therefore pair open artifacts with careful framing about scope, dated provider access, and the distinction between visible prosociality and resilient collective outcomes.

G. Selected Figures

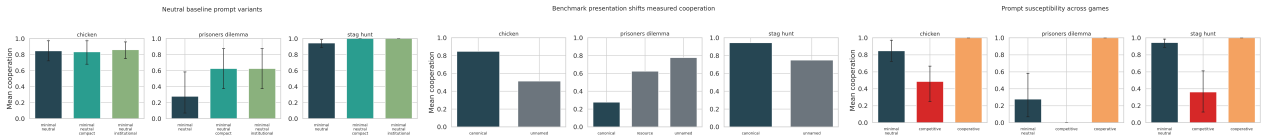

Figure 1. Selected repaired Part 1 figures: cross-game baseline comparison, benchmark-presentation comparison, and prompt-susceptibility comparison.

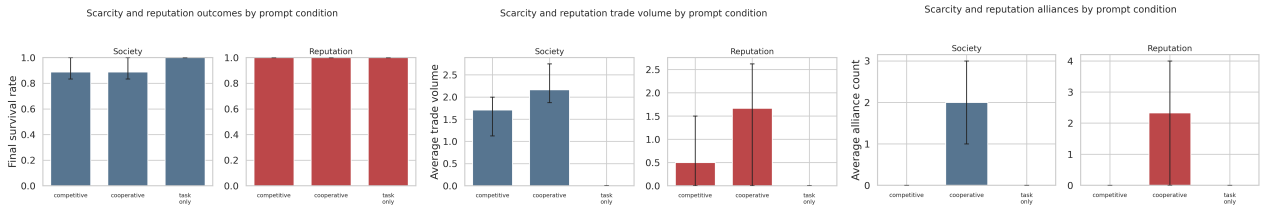

Figure 2. Completed corrected institutional figures for final survival, trade volume, and alliance count under scarcity and public reputation.

References

Core references currently used in the manuscript draft.

1. Nunzio Lore and Babak Heydari. *Strategic behavior of large language models and the role of game structure versus contextual framing*. Scientific Reports, 2024. URL: https://www.nature.com/articles/s41598-024-69032-z

2. Elif Akata et al. *Playing repeated games with large language models*. Nature Human Behaviour, 2025. URL: https://www.nature.com/articles/s41562-025-02172-y

3. Nicoló Fontana, Francesco Pierri, and Luca Maria Aiello. *Nicer Than Humans: How do Large Language Models Behave in the Prisoner's Dilemma?* arXiv, 2024. URL: https://arxiv.org/abs/2406.13605

4. Manuel Schmidt et al. *GPT-3.5 altruistic advice is sensitive to reciprocal concerns but not to strategic risk*. Scientific Reports, 2024. URL: https://www.nature.com/articles/s41598-024-73306-x

5. Valerio Capraro, Roberto Di Paolo, and Veronica Pizziol. *A publicly available benchmark for assessing large language models' ability to predict how humans balance self-interest and the interest of others*. Scientific Reports, 2025. URL: https://www.nature.com/articles/s41598-025-01715-7

6. Karthik Sreedhar, Alice Cai, Jenny Ma, Jeffrey V. Nickerson, and Lydia B. Chilton. *Simulating Cooperative Prosocial Behavior with Multi-Agent LLMs: Evidence and Mechanisms for AI Agents to Inform Policy Decisions*. arXiv, 2025. URL: https://arxiv.org/abs/2502.12504

7. Jinghua Piao et al. *AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents Advances Understanding of Human Behaviors and Society*. arXiv, 2025. URL: https://arxiv.org/abs/2502.08691

8. Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. *Generative Agents: Interactive Simulacra of Human Behavior*. arXiv, 2023. URL: https://arxiv.org/abs/2304.03442

9. Shaoguang Mao et al. *ALYMPICS: LLM Agents Meet Game Theory -- Exploring Strategic Decision-Making with AI Agents*. arXiv, 2024. URL: https://arxiv.org/abs/2311.03220

10. Kunlun Zhu et al. *MultiAgentBench: Evaluating the Collaboration and Competition of LLM agents*. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2025. URL: https://aclanthology.org/2025.acl-long.421/